

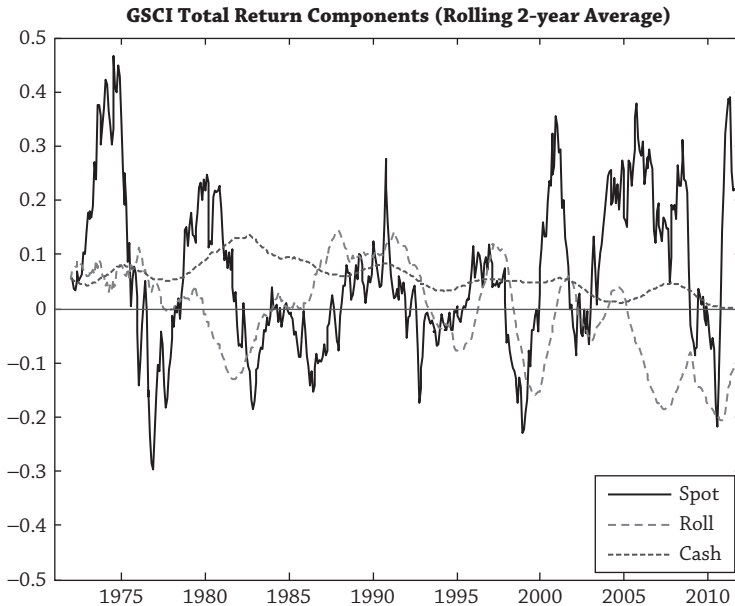


Figure 11.8

The cash and the roll components play important roles in the overall return of commodities, but the variance in the commodity future return is almost entirely due to movements in underlying spot markets. Thus all the risk is coming from spot markets. But to understand the expected returns, you need to understand all the factor drivers of all three spot, cash, and roll components.

Spot commodity markets are affected by production factors and investor behavior discussed in Section 5.1.

Interest rate factors determine the cash return (see chapter 9). Because you do not earn negative interest rates on collateral (at least not yet), the cash return is always positive.

The roll return depends on the slope of the futures curve. When the futures curve is upward sloping, called *contango*, there is a negative roll return.²⁴ When markets are in contango, an investor must replace the currently expiring futures contract with the next-maturity contract that is more expensive. This involves selling low and buying high, and so the roll loses money. A downward-sloping futures curve is called *backwardation*. When markets are *backwardated*, the investor sells the expiring futures at a high price and is able to buy the next-maturity contract at a low price. Selling high and buying low in backwardated futures markets makes money. But not always, because commodity markets can quickly change between backwardation and contango.

²⁴ See Mouakhar and Roberge (2010) for optimized roll strategies.